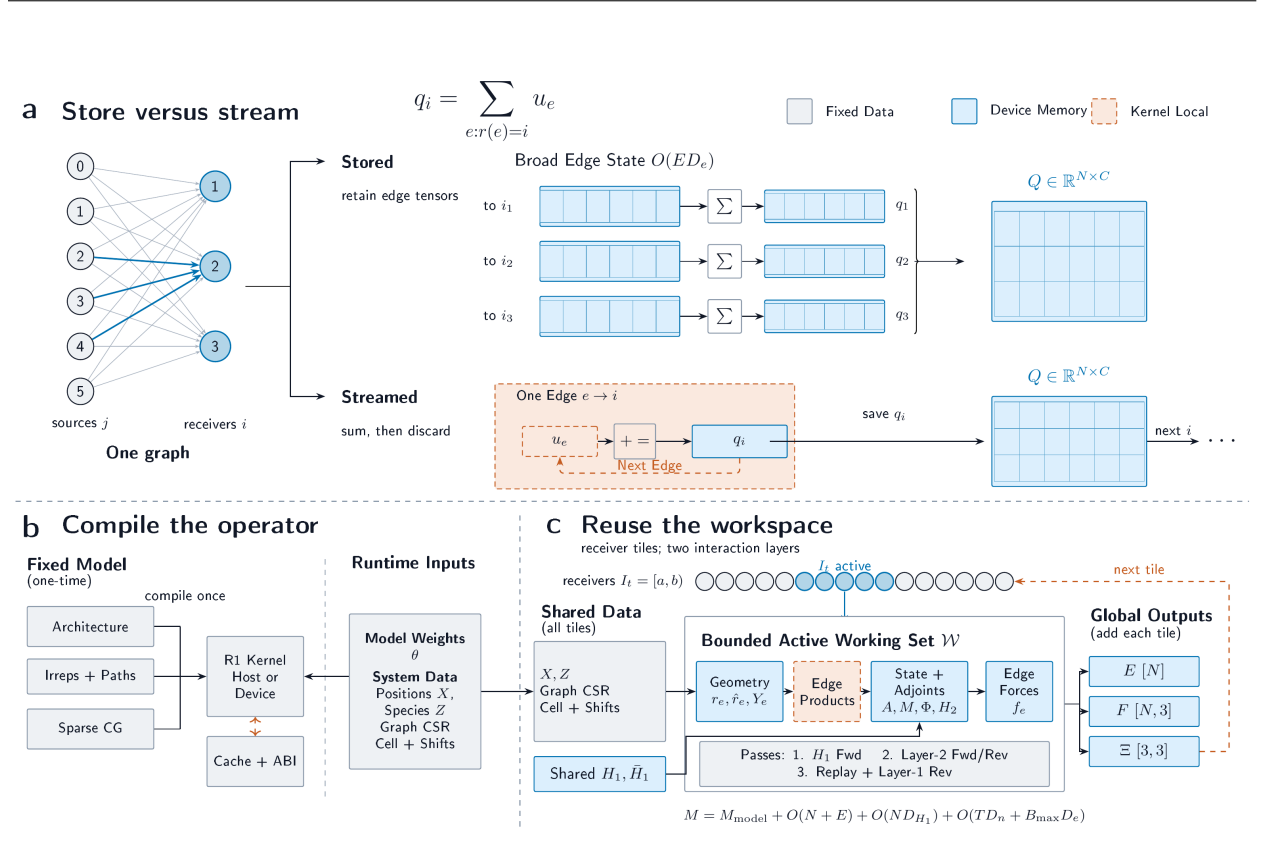

a Store versus stream
$q_i = \sum_{e:r(e)=i} u_e$
Fixed Data
Device Memory
Kernel Local
0
1
2
3
4
5
sources j
receivers i
One graph
Stored
retain edge tensors
Broad Edge State $O(ED_e)$
to i_1
to i_2
to i_3
q_1
q_2
q_3
$Q \in \mathbb{R}^{N \times C}$
Streamed
sum, then discard
One Edge $e \to i$
u_e
+ =
q_i
Next Edge
save q_i
$Q \in \mathbb{R}^{N \times C}$
next i
b Compile the operator
Fixed Model
(one-time)
compile once
Architecture
Irreps + Paths
Sparse CG
R1 Kernel Host or Device
Cache + ABI
Runtime Inputs
Model Weights θ
System Data
Positions X, Species Z
Graph CSR
Cell + Shifts
c Reuse the workspace
receiver tiles; two interaction layers
receivers $I_t = [a, b)$
I_t active
next tile
Shared Data
(all tiles)
X, Z
Graph CSR
Cell + Shifts
Shared $H_1, \bar{H}_1$
Bounded Active Working Set $\mathcal{W}$
Geometry $r_e, \hat{r}_e, Y_e$
Edge Products
State + Adjoints A, M, Φ, H_2
Edge Forces f_e
Passes: 1. H_1 Fwd 2. Layer-2 Fwd/Rev
3. Replay + Layer-1 Rev
Global Outputs
(add each tile)
E [N]
F [$N, 3$]
Ξ [3, 3]
$M = M_{\text{model}} + O(N + E) + O(ND_{H_1}) + O(TD_n + B_{\max} D_e)$